

# Simulating disease transmission

*(Part 1)*

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SCIENCE AND  
EDUCATION  
FOR  
SUSTAINABLE  
LIFE

# Overview

*This lecture will be in 3 parts:*

1. *General introduction to Epidemic Models*
2. *Simple Epidemic Models*
  - a) *Deterministic SIR model*
  - b) *Stochastic SIR model*
3. *Other Epidemic Models*
  - a) *SIS · SEIR · SIRS ...*

# The experts



Andrea Wilson



Javier Sanchez



Carsten T. Kirkeby



Cris Pooley



Stefan Widgren

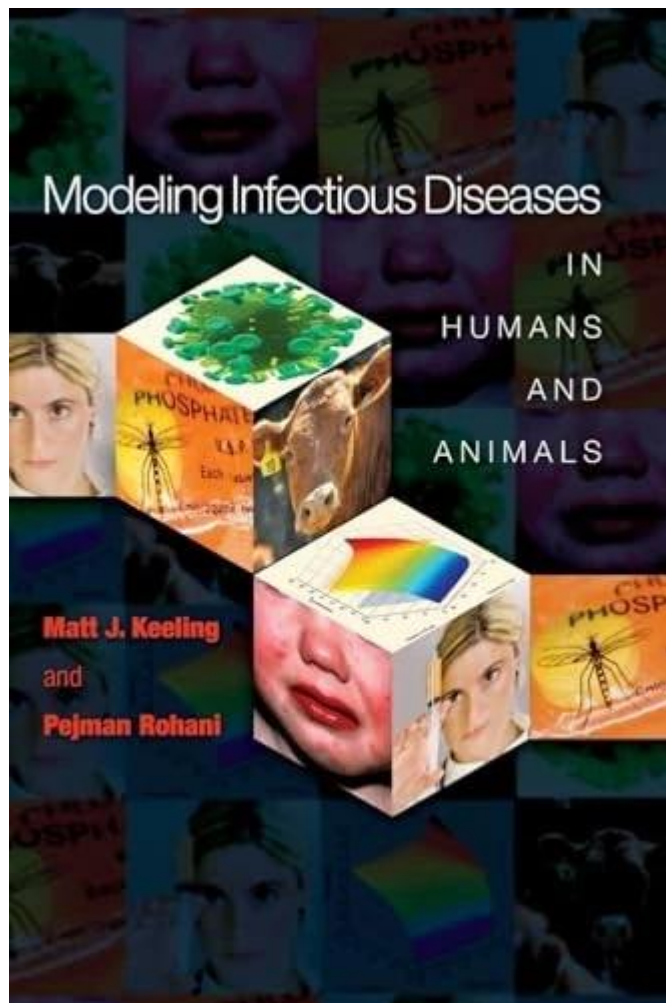


Tom Britton

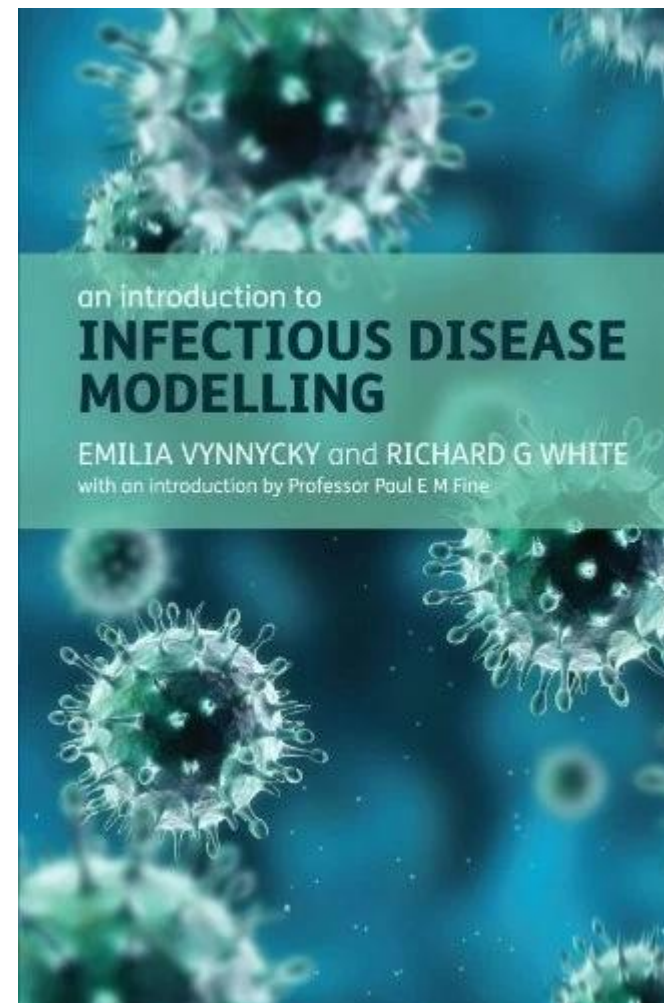


Svenja Woudstra

# Essential references



(Keeling & Rohani, 2011)



(Vynnycky & White, 2010)

# 1. General introduction to Epidemic models

# 1. General introduction to Epidemic models

- What models **CAN** do?

- **Prediction:** Predictive models can have great power in specific situations, guiding difficult policy decisions where a trade-off between two (or more) alternative control strategies exists.

- **Understanding:** Epidemic models provide epidemiologists with a ideal world in which individual factors can be examined in isolation and where every facet of the disease spread is recorded in perfect detail.

- What models **CANNOT** do?

- **Perfection:** A perfect epidemic model would need to account for variations in transmission with temperature and climate, capture the day-to-day movement and interaction of individuals, and encompass the variability in susceptibility due to genetic factors or past infections.

# 1. General introduction to Epidemic models

- What models **CANNOT** do?

***“We will never be able to predict the precise course of an epidemic, or which people will be infected. The best that we can hope for is models that provide confidence intervals on the epidemic behavior and determine the risk of infection for various groups of hosts.”***



Matt Keeling



Pejman Rohani

(Keeling & Rohani, 2011)

# 1. General introduction to Epidemic models

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# 1. General introduction to Epidemic models

- Modelling framework – Formulation

- **Deterministic models:**



Predicts a single outcome for a given set of parameters and initial conditions.

- **Stochastic models:**



Includes chance, so the same parameters can produce different outcomes.

- Modelling framework – Population representation

- **Compartmental models :**

Groups individuals into population states, such as susceptible, infected, or recovered.



- **Individual-based models :**

Represents each individual explicitly and tracks its state and characteristics over time.



# 1. General introduction to Epidemic models

|                      | <b>Deterministic</b>  | <b>Stochastic</b>  |
|----------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Fundamental question | What happens <b>on average</b> ?                                                                         | <b>What could happen</b> , and with what probability?                                                 |
| Output               | One trajectory<br>(Same result every run)                                                                | Distribution of trajectories<br>(Different result every run)                                          |
| Infection            | Expected inf. Rate<br>(e.g., $I = 0.17$ cows)                                                            | Discrete probabilistic event<br>(Transmission occur or does not')                                     |
| Extinction           | Difficult to represent naturally                                                                         | Occurs naturally                                                                                      |
| Best suited to       | Large populations<br>High number of infections                                                           | Small populations<br>Low number of infections<br>Introduction or elimination                          |
| Computational cost   | Low                                                                                                      | High                                                                                                  |

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# 1. General introduction to Epidemic models

Compartmental models



Individual-based models



|                      | Compartmental models                                                           | Individual-based models                                                               |
|----------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Fundamental question | What happens to <b>groups of individuals</b> ?                                 | What happens to <b>each individual</b> ?                                              |
| Representation       | Individuals grouped into compartments<br>(e.g. <b>S = 80</b> , <b>I = 20</b> ) | Each individual represented explicitly<br>(e.g. <b>Cow 1 = S</b> , <b>Cow 2 = I</b> ) |
| Heterogeneity        | Represented by adding groups/compartments                                      | Can be assigned directly to each individual                                           |
| Best suited to       | Population-level dynamics and relatively homogeneous populations               | Individual heterogeneity, specific contacts, spatial structure or complex behaviour   |
| Computational cost   | Low                                                                            | High                                                                                  |

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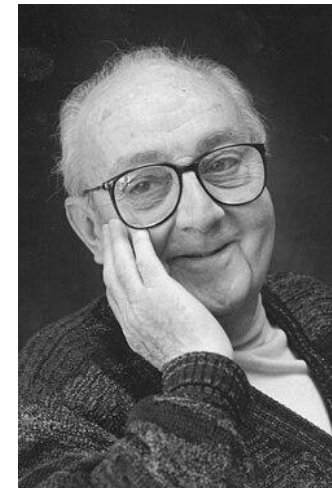
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# 1. General introduction to Epidemic models

*“All models are wrong, but some are useful.”*



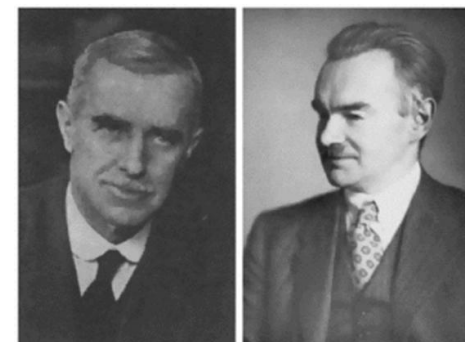
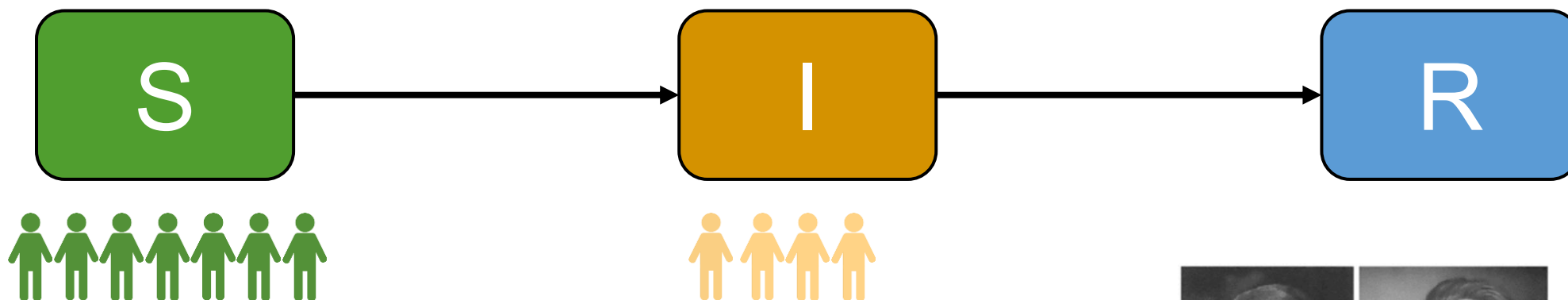
**Prof. George Box**

## ***2. Simple Epidemic Models***

## 2. Simple Epidemic Models

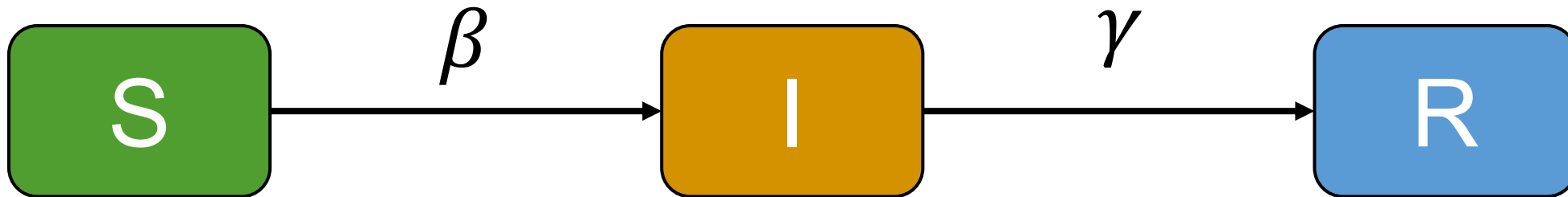


- The classical deterministic compartmental SIR model was introduced by Kermack & McKendrick (1927).



Kermack, W.O., & McKendrick, A.G., 1927. A contribution to the mathematical theory of epidemics. *Proc. R. Soc. A*. 115 (772): 700–721. <https://doi.org/10.1098/rspa.1927.0118>

## 2. Simple Epidemic Models



### Susceptible (S):

Number of individuals who can become infected  
(Population at risk)

### Infected (I):

Number of individuals who are infected and can transmit the disease.

### Recovered (R):

Number of individuals who have recovered and are not longer infectious.

### Transmission coefficient ( $\beta$ ):

*The rate at which susceptible individuals become infected.*

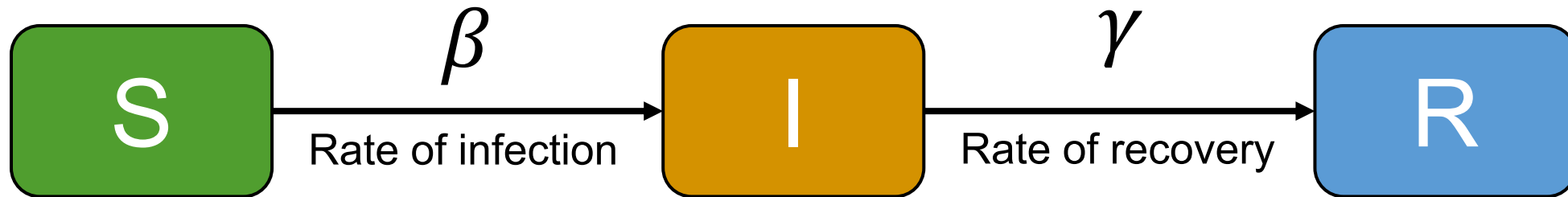
### Recovery rate ( $\gamma$ ):

*The rate at which infected individuals recover.*

### Days to recover ( $1/\gamma$ ):

*The inverse of the recover rate (individual/day) represent days to recover (day/ind).*

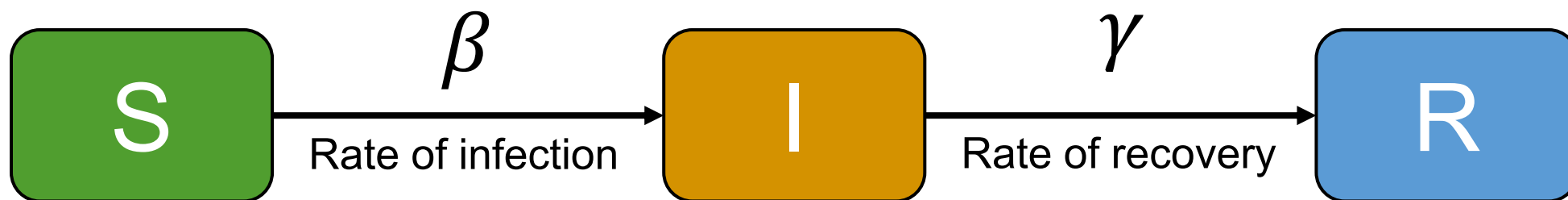
## 2. Simple Epidemic Models



### Assumptions

- The compartments represent the population ( $S + I + R = N$ )
- No birds or immigrations ( $N$  is constant)
- People are well mixed
- Infection rate is fixed ( $\beta$  is constant)
- Fixed recover times ( $\gamma$  is constant)
- Recovery gives total immunity

## 2. Simple Epidemic Models



$$S + I + R = N$$

$$\dot{S} = -\beta IS$$

→

$$\frac{dS}{dt} = -\beta IS$$

Rate of change of S (dS) versus time dt

$$\dot{I} = +\beta IS - \gamma I$$

→

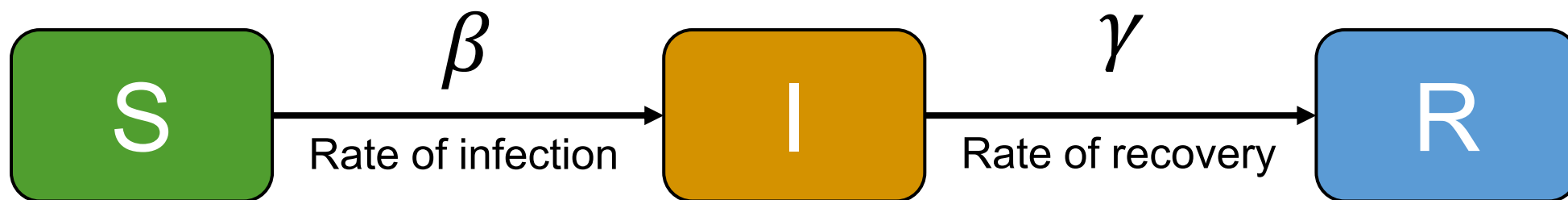
$$\frac{dI}{dt} = +\beta IS - \gamma I$$

$$\dot{R} = +\gamma I$$

→

$$\frac{dR}{dt} = +\gamma I$$

## 2. Simple Epidemic Models



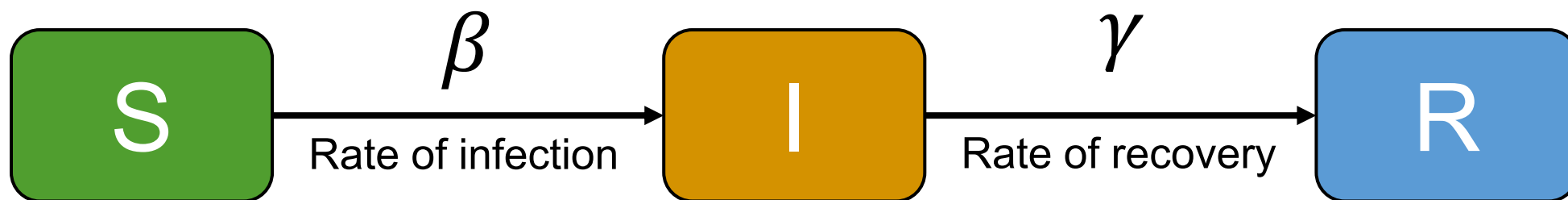
$$S + I + R = N$$

$$\frac{dS}{dt} = -\beta IS \quad \rightarrow \quad S_{t+1} - S_t = -(\beta I_t S_t)dt$$

$$\frac{dI}{dt} = +\beta IS - \gamma I \quad \rightarrow \quad I_{t+1} - I_t = +(\beta I_t S_t - \gamma I_t)dt$$

$$\frac{dR}{dt} = +\gamma I \quad \rightarrow \quad R_{t+1} - R_t = +(\gamma I_t)dt$$

## 2. Simple Epidemic Models



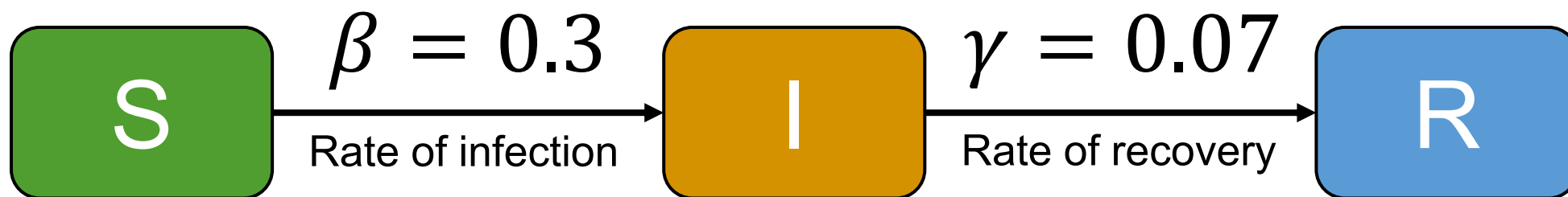
$$S + I + R = N$$

$$\frac{dS}{dt} = -\beta IS \quad \rightarrow \quad S_{t+1} = S_t - (\beta I_t S_t)dt$$

$$\frac{dI}{dt} = +\beta IS - \gamma I \quad \rightarrow \quad I_{t+1} = I_t + (\beta I_t S_t - \gamma I_t)dt$$

$$\frac{dR}{dt} = +\gamma I \quad \rightarrow \quad R_{t+1} = R_t + (\gamma I_t)dt$$

## 2. Simple Epidemic Models

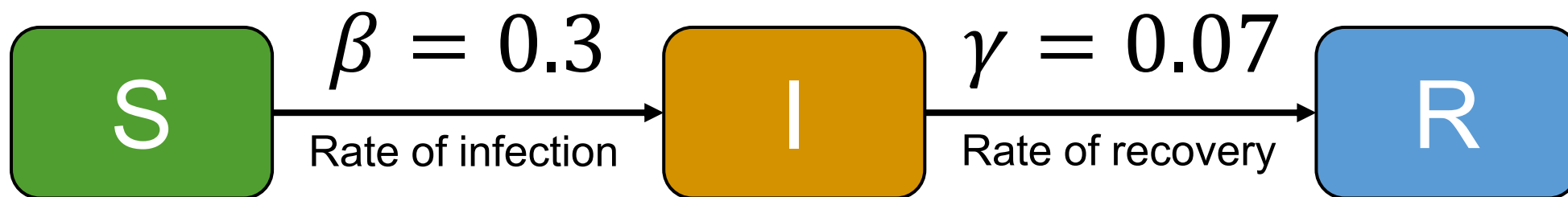


| Time | Susceptible | Infected | Recovered |
|------|-------------|----------|-----------|
| 1    | 0.999       | 0.001    | 0.000     |

$$S_{t+1} = S_t - (\beta I_t S_t) dt$$

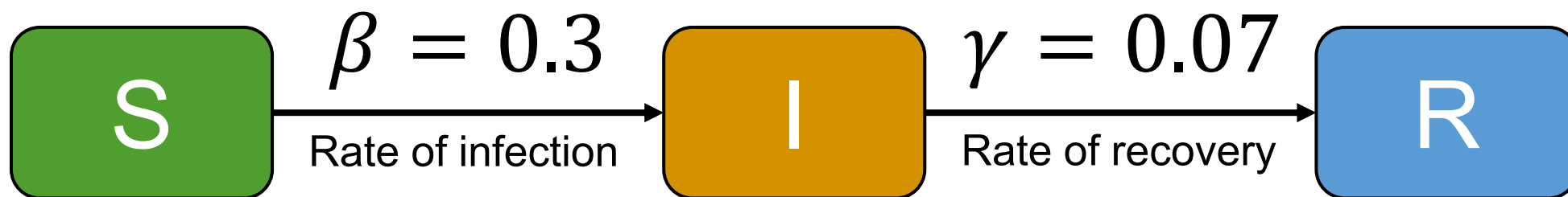
|   |           |  |  |
|---|-----------|--|--|
| 2 | $S_{t+1}$ |  |  |
|---|-----------|--|--|

## 2. Simple Epidemic Models



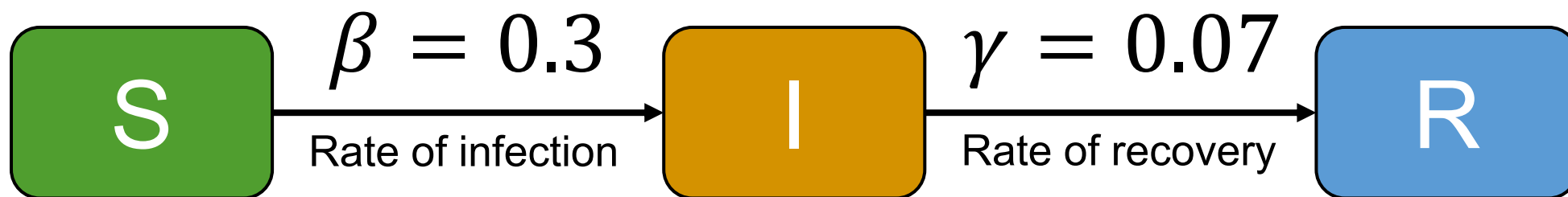
| Time                                          | Susceptible        | Infected | Recovered |
|-----------------------------------------------|--------------------|----------|-----------|
| 1                                             | $s_t = 0.999$      | 0.001    | 0.000     |
| $s_{t+1} = 0.999 - (0.3 * 0.001 * 0.999) * 1$ |                    |          |           |
| 2                                             | $s_{t+1} = 0.9987$ |          |           |

## 2. Simple Epidemic Models



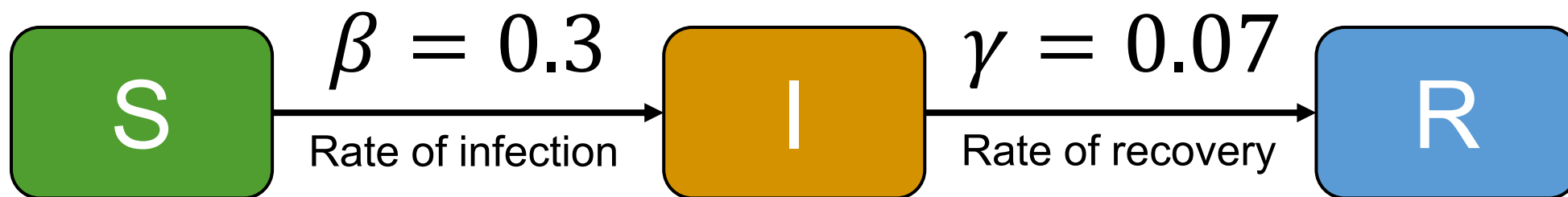
| Time                                              | Susceptible        | Infected  | Recovered |
|---------------------------------------------------|--------------------|-----------|-----------|
| 1                                                 | $S_t = 0.999$      | 0.001     | 0.000     |
| $I_{t+1} = I_t + (\beta I_t S_t - \gamma I_t) dt$ |                    |           |           |
| 2                                                 | $S_{t+1} = 0.9987$ | $I_{t+1}$ |           |

## 2. Simple Epidemic Models



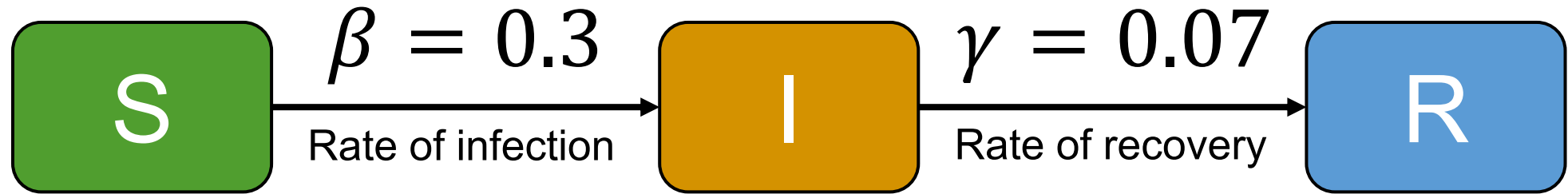
| Time                                                         | Susceptible        | Infected           | Recovered |
|--------------------------------------------------------------|--------------------|--------------------|-----------|
| 1                                                            | $S_t = 0.999$      | $I_t = 0.001$      | 0.000     |
| $I_{t+1} = 0.001 + (0.3 * 0.001 * 0.999 - 0.07 * 0.001) * 1$ |                    |                    |           |
| 2                                                            | $S_{t+1} = 0.9987$ | $I_{t+1} = 0.0012$ |           |

## 2. Simple Epidemic Models



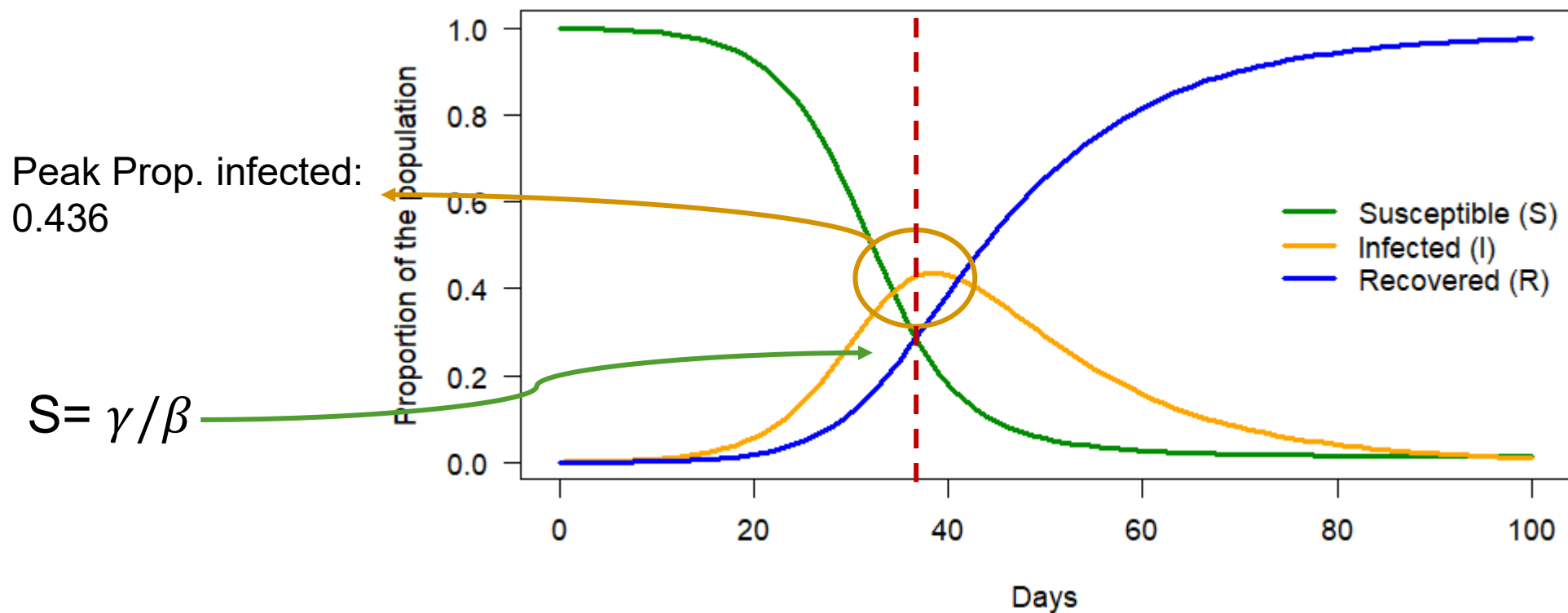
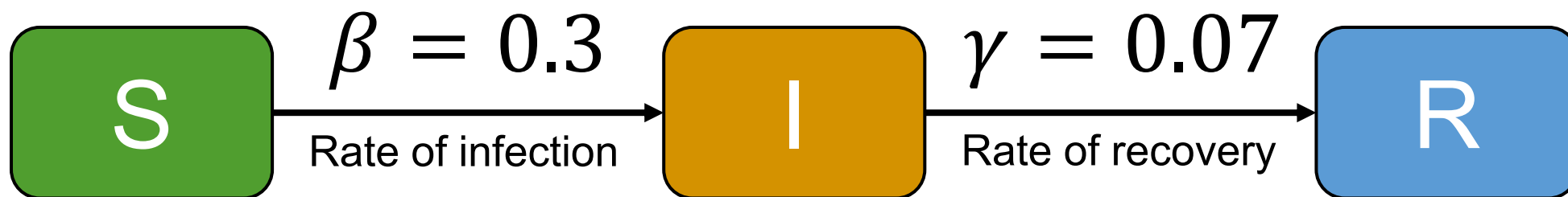
| Time                             | Susceptible        | Infected           | Recovered |
|----------------------------------|--------------------|--------------------|-----------|
| 1                                | $S_t = 0.999$      | $I_t = 0.001$      | 0.000     |
| $R_{t+1} = R_t + (\gamma I_t)dt$ |                    |                    |           |
| 2                                | $S_{t+1} = 0.9987$ | $I_{t+1} = 0.0012$ | $R_{t+1}$ |

## 2. Simple Epidemic Models

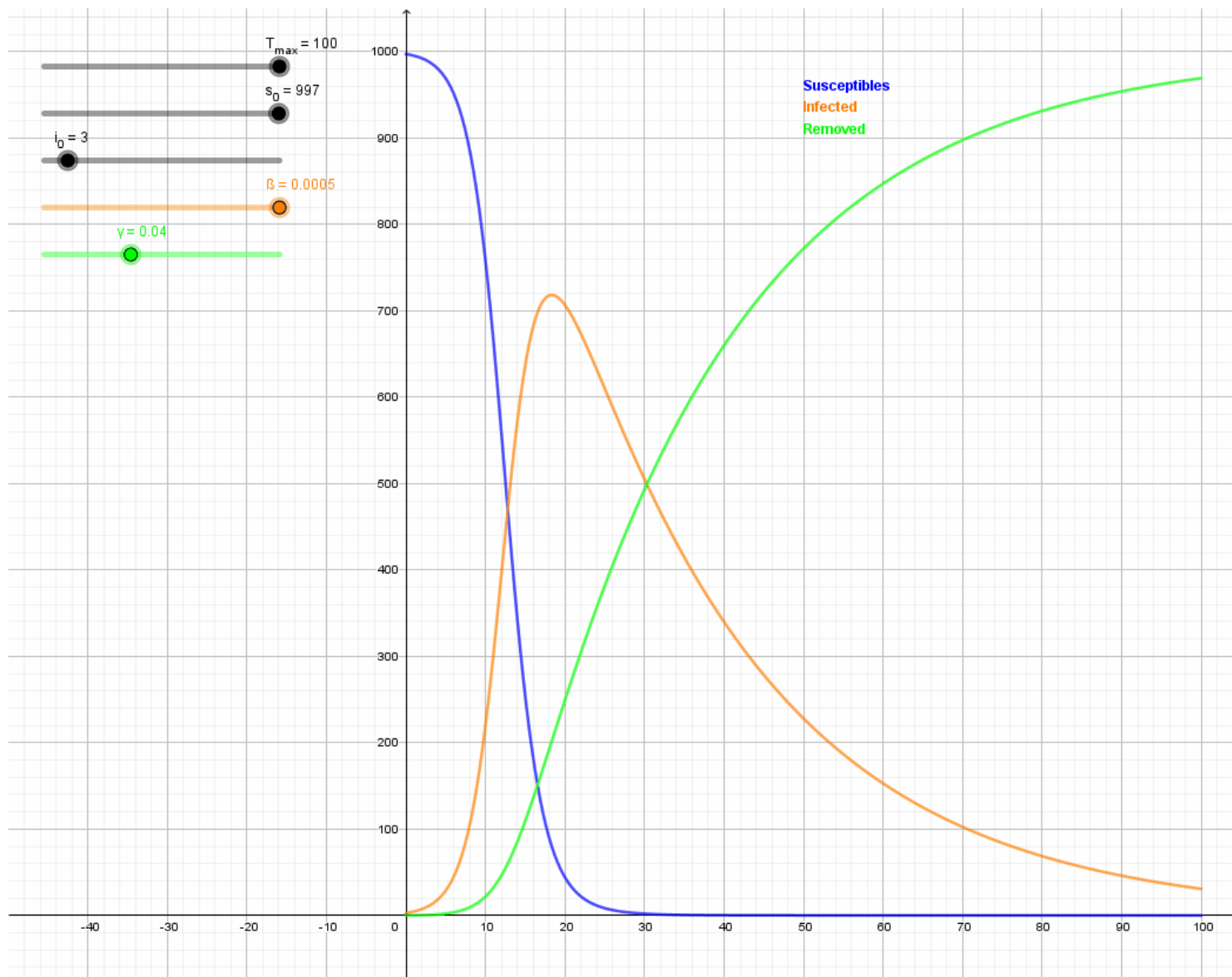


| Time                                   | Susceptible        | Infected           | Recovered         |
|----------------------------------------|--------------------|--------------------|-------------------|
| 1                                      | $S_t = 0.999$      | $I_t = 0.001$      | $R_t = 0.000$     |
| $R_{t+1} = 0.000 + (0.07 * 0.001) * 1$ |                    |                    |                   |
| 2                                      | $S_{t+1} = 0.9987$ | $I_{t+1} = 0.0012$ | $R_{t+1} = 7e-05$ |

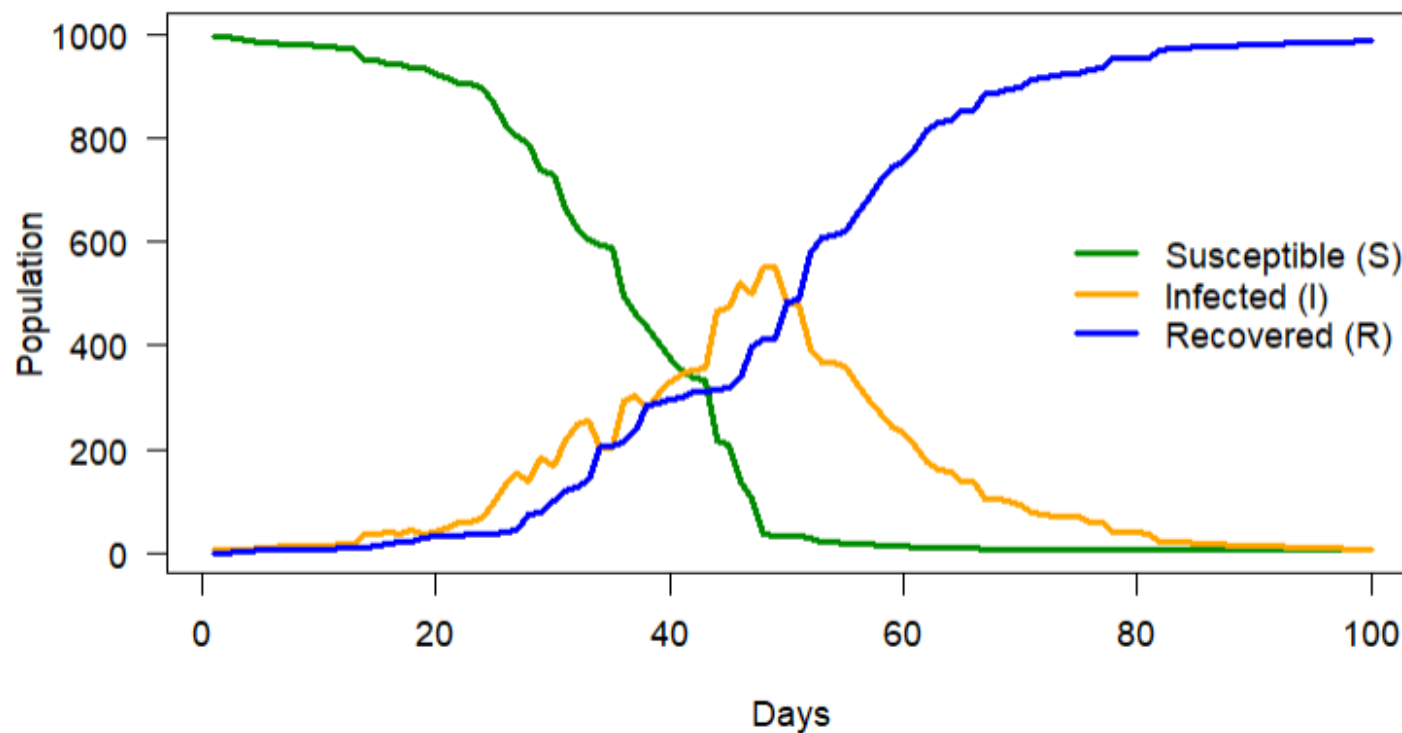
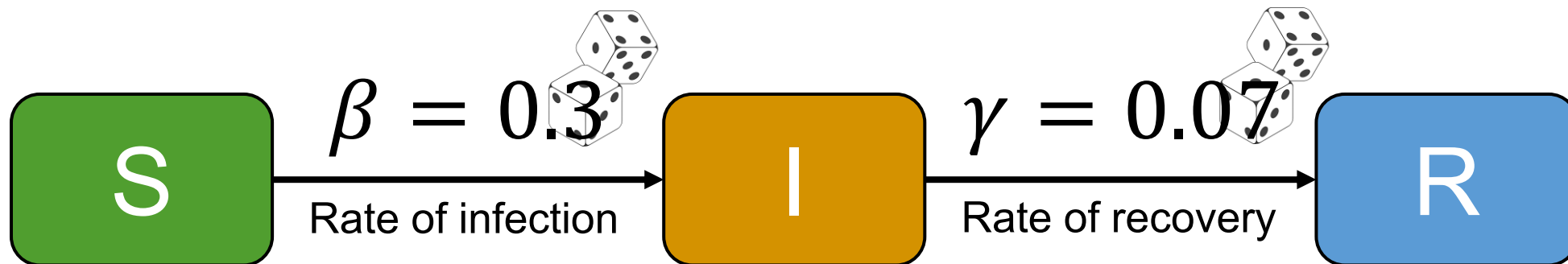
## 2. Simple Epidemic Models



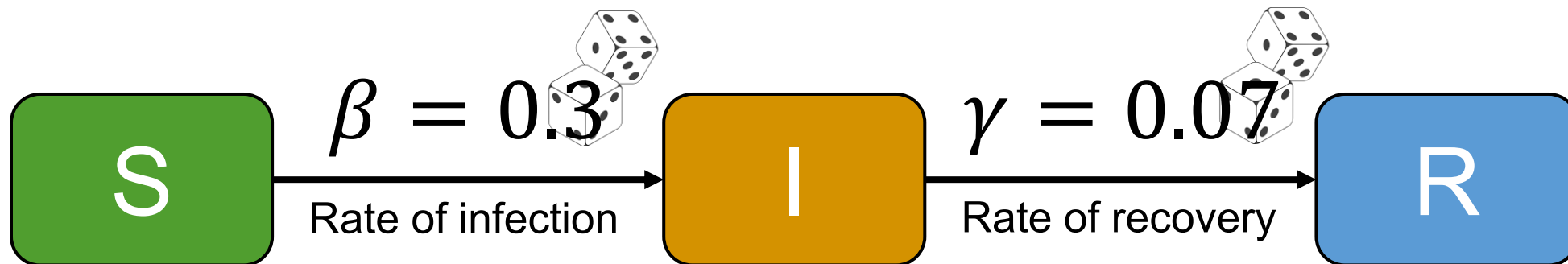
## 2. Simple Epidemic Models



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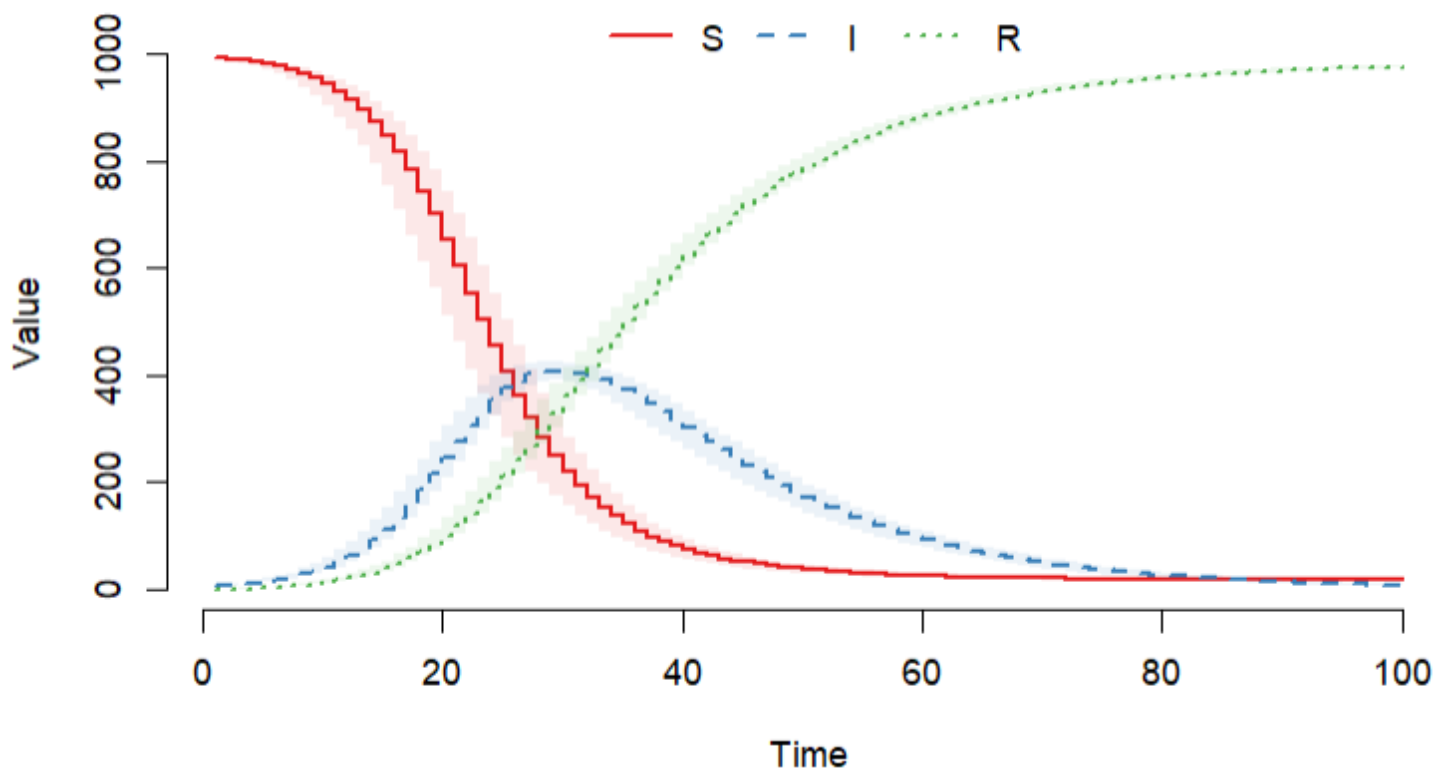
## 2. Simple Epidemic Models



SimInf



Stefan Widgren

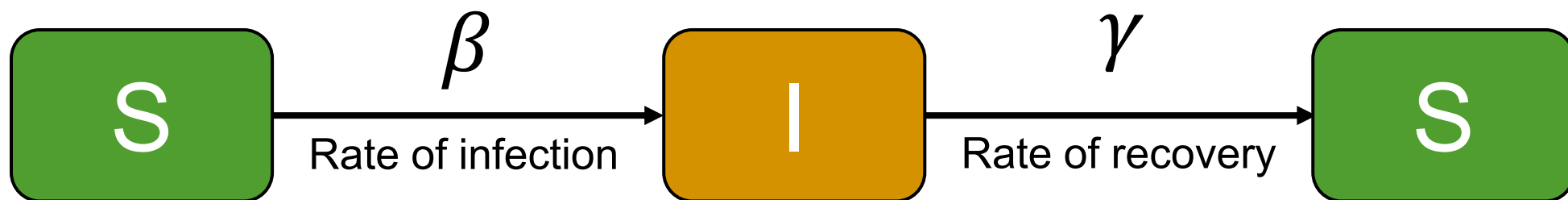


# ***3. Other Epidemic Models***

### 3. *Other Epidemic Models*

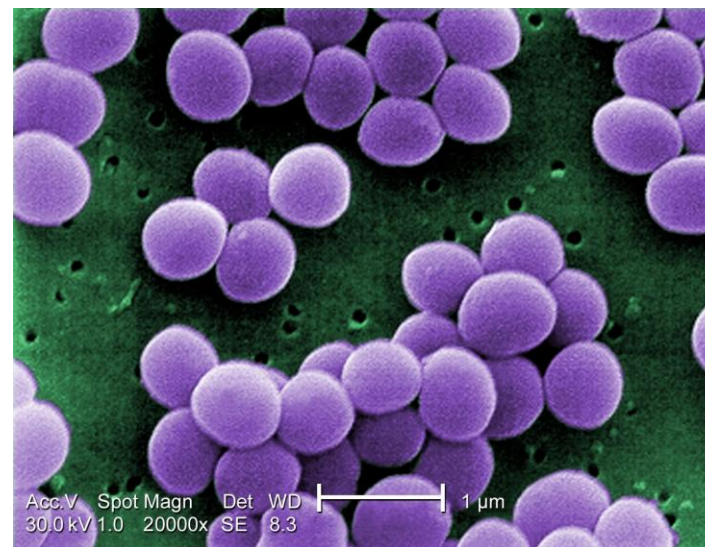
- Susceptible-Infected-Recovered (SIR)
- Model without immunity (SIS)
  - Susceptible-Infected-Susceptible
- Model waning immunity (SIRS)
  - Susceptible-Infected-Recovered-Susceptible
- Model with latent period (SEIR)
  - Susceptible-Exposed-Infected-Recovered

### 3. Other Epidemic Models

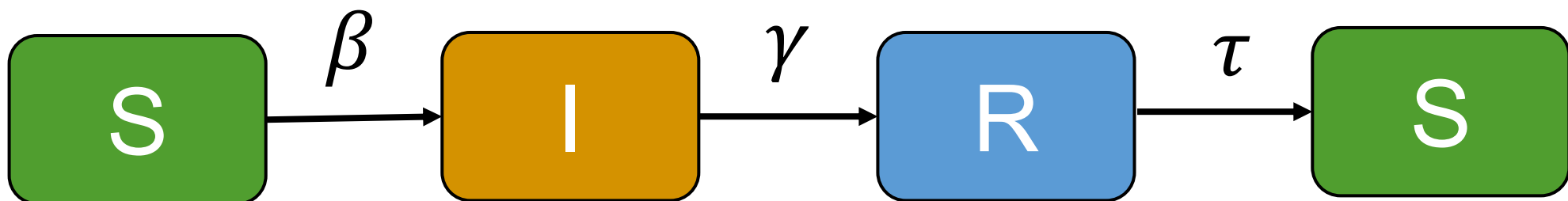


$$S + I = N$$

$$\begin{aligned}\dot{S} &= -\beta IS + \gamma I \\ \dot{I} &= +\beta IS - \gamma I\end{aligned}$$



### 3. Other Epidemic Models



$$S + I + R = N$$

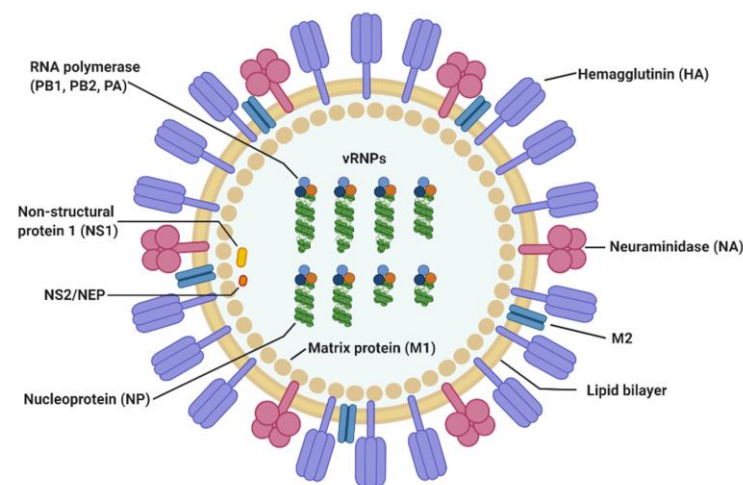
$$\dot{S} = -\beta IS + \tau R$$

$$\dot{I} = +\beta IS - \gamma I$$

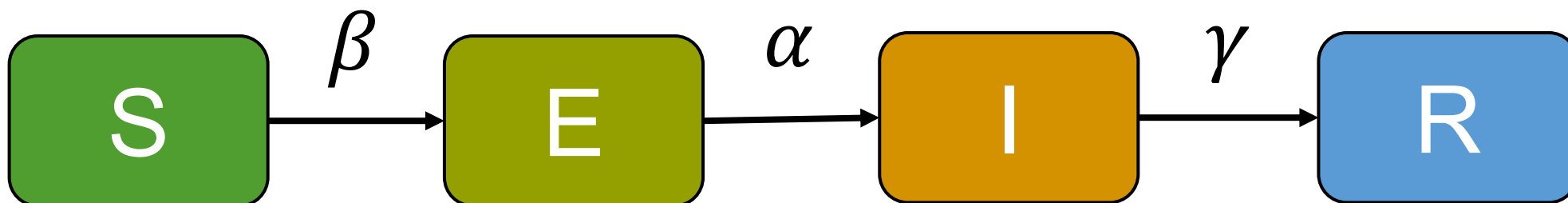
$$\dot{R} = \gamma I - \tau R$$

#### Exposed:

Incubation period during which individuals have already been infected but are not yet infectious



### 3. Other Epidemic Models

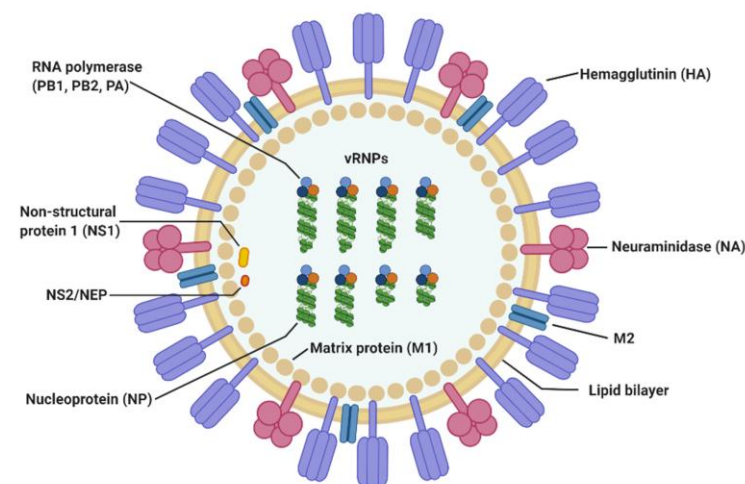


$$S + E + I + R = N$$

$$\begin{aligned}\dot{S} &= -\beta IS \\ \dot{E} &= +\beta IS - \alpha E \\ \dot{I} &= \alpha E - \gamma I \\ \dot{R} &= +\gamma I\end{aligned}$$

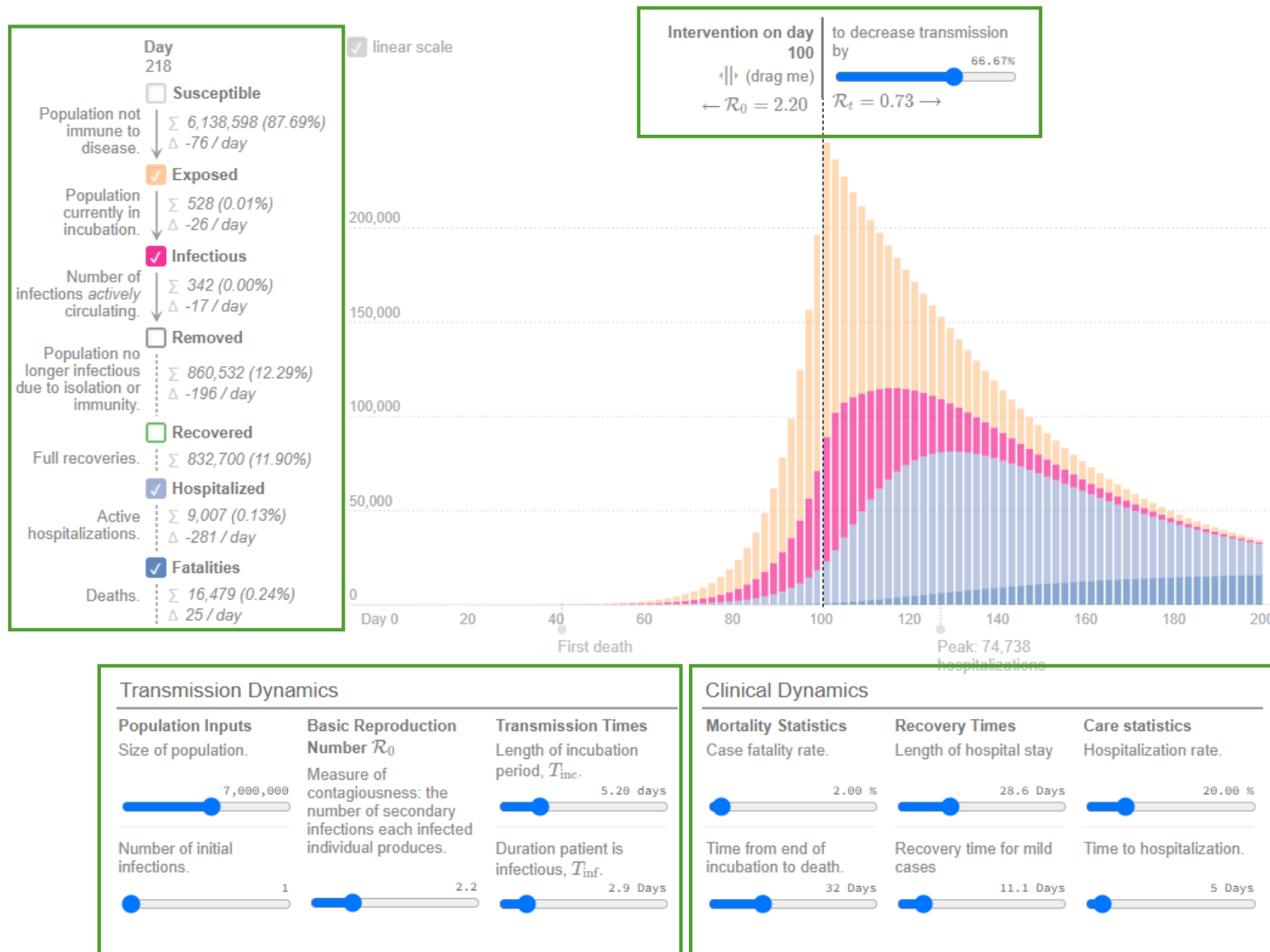
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# 3. Other Epidemic Models

## Epidemic Calculator



Gabriel Goh

- *Essential*

- Keeling, M.J., Rohani, P., 2011. Modeling infectious diseases in humans and animals, Modeling Infectious Diseases in Humans and Animals. [https://doi.org/10.1016/s1473-3099\(08\)70147-6](https://doi.org/10.1016/s1473-3099(08)70147-6)
- Vynnycky, E., White, R.G., 2010. An introduction to infectious disease modelling. Oxford University Press, New York. <https://doi.org/10.1093/eurpub/ckq163>

- *Further*

- Widgren, S., Bauer, P., Eriksson, R., & Engblom, S. 2019. SimInf: An R Package for Data-Driven Stochastic Disease Spread Simulations. *Journal of Statistical Software*. <https://doi.org/10.18637/jss.v091.i12>

hands  
on





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